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Lots of Hopf Algebras

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The purpose of this paper is to give a "universal" construction of Hopf algebras over any commutative ring. The mod-p Steenrod algebras are all examples of this construction; other examples of Hopf algebras related to the mod-p Steenrod algebras are also produced. Then we give an argument to produce deformations of these Hopf algebras via "twisting." The techniques of this paper are, in a way, "scheme-theoretic," producing many rather complicated Hopf algebras, only a few of which are studied in any detail here; however, we hope to return to the study of some of these Hopf algebras in future work.

1. DEFORMATIONS OF HOPF ALGEBRAS

Suppose that k is a commutative ring with 1, and that $(A, \iota, \epsilon, \Delta, S)$ is a Hopf algebra over k. Here, ι is the unit, ϵ the counit, μ the multiplication, Δ the comultiplication, and S the antipode of A. An *n-parameter deformation* of A is a topological Hopf algebra $(A_{\vec{h}}, \iota, \mu_{\vec{h}}, \epsilon, \Delta_{\vec{h}}, S_{\vec{h}})$ over the power series ring $k[[\vec{h}]]$ (where $\vec{h} = (h_1, \dots, h_n)$ is a set of n commuting indeterminates) such that

(a) $A_{\vec{h}}$ is isomorphic to $A[[\vec{h}]]$ as a topological $k[[\vec{h}]]$-module;
(b) $\mu_{\vec{h}} \equiv \mu \bmod \vec{h}$, $\Delta_{\vec{h}} \equiv \Delta \bmod \vec{h}$.

Two deformations $A_{\vec{h}}$ and $B_{\vec{h}}$ of the same Hopf algebra A are *equivalent* if there is an isomorphism $f_{\vec{h}}\colon A_{\vec{h}} \to B_{\vec{h}}$ of Hopf algebras over $k[[\vec{h}]]$ which is the identity mod $\vec{h}$.

(We are using the notation and definitions of [2].)

We are assuming that the unit and counit of $A_{\vec{h}}$ are obtained by simply extending those of A, so that there will be no difference in the notation for these maps, whether considered on A or on $A_{\vec{h}}$. Note that

$$\Delta_{\vec{h}}\colon A[[\vec{h}]] \to A[[\vec{h}]] \,\hat{\otimes}_{k[[\vec{h}]]}\, A[[\vec{h}]]$$

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and that

$$\mu_{\vec{h}}\colon A[[\vec{h}]] \mathbin{\hat{\otimes}}_{k[[\vec{h}]]} A[[\vec{h}]] \to A[[\vec{h}]];$$

the completion indicated by "$\hat{\otimes}$" is the $\vec{h}$-adic completion. We will also make the natural identification of $k[[\vec{h}]]$-modules

$$A[[\vec{h}]] \mathbin{\hat{\otimes}}_{k[[\vec{h}]]} A[[\vec{h}]] \cong (A \otimes_k A)[[\vec{h}]]$$

without further comment.

The *trivial* n-parameter deformation of A is the deformation where $\mu_{\vec{h}}$, $\Delta_{\vec{h}}$, and $S_{\vec{h}}$ are obtained by extending μ, Δ, and S $k[[\vec{h}]]$-linearly to $A[[\vec{h}]]$. In other words,

$$\mu_{\vec{h}}\left(\sum_{\vec{\alpha}} a_{\vec{\alpha}} \vec{h}^{\vec{\alpha}} \otimes \sum_{\vec{\alpha}} b_{\vec{\alpha}} \vec{h}^{\vec{\alpha}} \right) = \sum_{\vec{\alpha}} \sum_{\vec{\beta}} \mu\left(a_{\vec{\alpha}} \otimes b_{\vec{\beta}}\right) \vec{h}^{\vec{\alpha}+\vec{\beta}},$$

$$\Delta_{\vec{h}}\left(\sum_{\vec{\alpha}} a_{\vec{\alpha}} \vec{h}^{\vec{\alpha}} \right) = \sum_{\vec{\alpha}} \Delta(a_{\vec{\alpha}}) \hat{h}^{\vec{\alpha}}$$

and

$$S\left(\sum_{\vec{\alpha}} a_{\vec{\alpha}} \vec{h}^{\vec{\alpha}} \right) = \sum_{\vec{\alpha}} S(a_{\vec{\alpha}}) \vec{h}^{\vec{\alpha}}.$$

Here and elsewhere, $\vec{\alpha} = (a_1, \dots, a_n)$ is a multi-index of nonnegative integers and $\vec{h}^{\vec{\alpha}} = h_1^{\alpha_1} \cdots h_n^{\alpha_n}$.

If B is any (topological) Hopf algebra (over a commutative (topological) ring), with comultiplication Δ_B and antipode S_B, define the set of *group-like* elements of B as

$$G(B) = \left\{ b \in B - \{0\} \mid \Delta_B(b) = b \otimes b \right\}.$$

We know that (see, e.g., [8])

(a) $G(B)$ is a monoid with respect to the multiplication in B, with identity the unit 1 of B; in particular, for all $b \in B$, $bS_B(b) = \epsilon(b)1$.

(b) For every $b \in B$, we have

$$\Delta(S_B(b)) = S_B(b) \otimes S_B(b),$$

so $S_B(b) \in G(B)$ if $S_B(b)$ does not equal zero.

(c) If $b \in G(B)$ and $\epsilon(b) = 1$, then $S_B(b)$ is the inverse of b in the monoid $G(B)$.

The proof the following lemma is immediate.

LEMMA 1.0.1. *Let B be a (topological) Hopf algebra over a commutative (topological) ring R, with antipode S_B and comultiplication Δ_B. We denote the multiplication in B by juxtaposition.*

(i) *If $a, b \in G(B)$, then $\Delta_B(a-b) = (a-b)\otimes a + b \otimes (a-b)$.*

(ii) *If $a, b \in G(B)$ are such that $\epsilon(a) = \epsilon(b) = 1$, then $S_B(a-b) = S_B(a)(b-a)S_B(b)$.*

(iii) *Suppose that e, $\tilde{e}$, g, and $\tilde{g}$ are grouplike elements of B. Let z and $\tilde{z}$ in B satisfy*

$$\Delta_B(z) = z \otimes g + g \otimes z$$

and

$$\Delta_B(\tilde{z}) = \tilde{z} \otimes \tilde{g} + \tilde{g} \otimes \tilde{z}.$$

Then

$$\begin{aligned}\Delta_B(ze - \tilde{z}\tilde{e}) = {} & (ze - \tilde{z}\tilde{e}) \otimes ge + ge \otimes (ze - \tilde{z}\tilde{e}) \\ & + \tilde{z}\tilde{e} \otimes (ge - \tilde{g}\tilde{e}) + (ge - \tilde{g}\tilde{e}) \otimes \tilde{z}\tilde{e}.\end{aligned}$$

(iv) *Suppose that z, $\tilde{z}$, e, $\tilde{e}$, g, and $\tilde{g}$ are as in* (iii) *and that, furthermore, $\epsilon(z) = \epsilon(\tilde{z}) = 0$ and $\epsilon(e) = \epsilon(\tilde{e}) = \epsilon(g) = \epsilon(\tilde{g}) = 1$. Then*

$$\begin{aligned}S_B(ze - \tilde{z}\tilde{e}) = {} & -S_B(ge)(ze - \tilde{z}\tilde{e})S_B(ge) - S_B(\tilde{z}\tilde{e})(ge - \tilde{g}\tilde{e})S_B(ge) \\ & - S_B(ge)(\tilde{g}\tilde{e} - ge)S_B(\tilde{g}\tilde{e})\tilde{z}\tilde{e}S_B(ge).\end{aligned}$$

Let t be an indeterminate. If B is any (topological) Hopf algebra over a commutative ring, with coalgebra map Δ_B, then a *curve over b_0* in B is an element $\gamma(t) = \sum_{i=0}^{\infty} b_i t^i$ of $B[[t]]$ such that $\Delta_B(b_n) = \sum_{j=0}^{n} b_j \otimes b_{n-j}$ for every n. A curve over 1 is also called a *divided power sequence* in B. The set of curves in B is naturally a monoid under ordinary power series multiplication.

Notice that if A_0 is the trivial one-parameter deformation of A, then the monoid of curves in A is naturally isomorphic with the monoid of grouplike elements in A_0. The isomorphism is obtained by setting the "curve parameter" t equal to the "deformation parameter" h.

Similarly, one may define *multiparameter curves over b_0*, where B is any Hopf algebra over a commutative ring, and $b_0 \in B$.

Given one grouplike element of a deformation of A, we may form many other as follows.

LEMMA 1.0.2. *Suppose t is an indeterminate (different from the indeterminates $h_1, \ldots, h_n$, and commuting with these). Let $p(\vec{h}) \in k[[\vec{h}]]$ be a power series with no constant term. Then*

(i) *The function $ev_p\colon A_{\vec{h}}[[t]] \to A_{\vec{h}}$ defined by $ev_p(\xi(t)) = \xi(p(\vec{h}))$ is a $k[[\vec{h}]]$-algebra homomorphism.*

(ii) *If $A_{\vec{h}}[[t]]$ is the trivial one-parameter deformation of the Hopf algebra $A_{\vec{h}}$, then ev_p is a homomorphism of topological Hopf algebras over $k[[\vec{h}]]$.*

Proof. (i) follows from the fact that $A_{\vec{h}}$ is (isomorphic to), as a topological $k[[\vec{h}]]$-module, $A[[\vec{h}]]$. Thus it makes sense to speak of the $\vec{h}$-adic topology on $A_{\vec{h}}$, and furthermore, we have that $A_{\vec{h}}$ is complete with respect to this topology. Since $k[[\vec{h}]]$ is also a central subalgebra of $A_{\vec{h}}$, we may evaluate power series over $A_{\vec{h}}$ at elements of the ideal $(h_1, \ldots, h_n)$ of $k[[\vec{h}]]$, and this is an algebra homomorphism.

Similarly, (ii) follows from the definition of the trivial deformation, and the fact that we are tensoring over $k[[\vec{h}]]$ (at least), and the completeness of everything in sight with respect to the $\vec{h}$-adic topology.

A special case of the lemma above is

COROLLARY 1.0.3. *Let $A_{\vec{h},0}$ be the trivial one-parameter deformation of $A_{\vec{h}}$. Then if $p(\vec{h})$ is as above,*

$$ev_p\colon G(A_{\vec{h},0}) \to G(A_{\vec{h}})$$

is a homomorphism of monoids.

Suppose that $a, b \in G(A_{\vec{h}})$. Write $a - b$ as a power series in $\vec{h}$:

$$a - b = \sum_{\vec{\alpha}} x_{\vec{\alpha}}(a, b)\vec{h}^{\vec{\alpha}},$$

where $x_{\vec{\alpha}}(a, b) \in A$ for every $\vec{\alpha}$. Define $I(a, b)$ to be the two-sided ideal of A generated by the $x_{\vec{\alpha}}(a, b)$.

THEOREM 1.0.4. *Let $a, b \in G(A_0)$, where A_0 is the trivial n-parameter deformation of A; suppose also that $\epsilon(a) = e(b) = 1$. Then $I(a, b)$ is a Hopf ideal in A.*

Proof. Let $a = \sum_{\vec{\alpha}} a_{\vec{\alpha}}\vec{h}^{\vec{\alpha}}$ and $b = \sum_{\vec{\alpha}} b_{\vec{\alpha}}\vec{h}^{\vec{\alpha}}$, where $a_{\vec{\alpha}}$ and $b_{\vec{\alpha}}$ are in A for every $\vec{\alpha}$.

Writing $a - b = \sum_{\vec{\alpha}} x_{\vec{\alpha}}(a, b)\vec{h}^{\vec{\alpha}}$, the equation

$$\Delta_{\vec{h}}(a - b) = (a - b) \otimes a + b \otimes (a - b)$$

becomes

$$\begin{aligned}
&\sum_{\vec{\alpha}} \Delta\big(x_{\vec{\alpha}}(a,b)\big)\vec{h}^{\vec{\alpha}} \\
&\qquad = \left(\sum_{\vec{\alpha}} x_{\vec{\alpha}}(a,b)\vec{h}^{\vec{\alpha}}\right) \otimes \sum_{\vec{\alpha}} a_{\vec{\alpha}}\vec{h}^{\vec{\alpha}} + \sum_{\vec{\alpha}} b_{\vec{\alpha}}\vec{h}^{\vec{\alpha}} \otimes \left(\sum_{\vec{\alpha}} x_{\vec{\alpha}}(a,b)\vec{h}^{\vec{\alpha}}\right) \\
&\qquad = \sum_{\vec{\alpha}} \left(\sum_{\vec{\gamma}+\vec{\beta}=\vec{\alpha}} x_{\vec{\gamma}}(a,b) \otimes a_{\vec{\beta}} + b_{\vec{\gamma}} \otimes x_{\vec{\beta}}(a,b)\right)\vec{h}^{\vec{\alpha}};
\end{aligned}$$

this implies that, for every $\vec{\alpha}$,

$$\Delta\big(x_{\vec{\alpha}}(a,b)\big) = \sum_{\vec{\gamma}+\vec{\beta}=\vec{\alpha}} x_{\vec{\gamma}}(a,b) \otimes a_{\vec{\beta}} + b_{\vec{\gamma}} \otimes x_{\vec{\beta}}(a,b),$$

which implies that $I(a,b)$ is a coideal in A.

Now, let S be the antipode for A (and by extension for A_0). Writing out the equation of Lemma 1.0.1(ii) above, we see also that $I(a,b)$ is invariant under S:

$$\begin{aligned}
\sum_{\vec{\alpha}} S\big(x_{\vec{\alpha}}(a,b)\big)h^{\vec{\alpha}} &= S(a-b) = S(a)(b-a)S(b) \\
&= -\left(\sum_{\vec{\alpha}} S(a_{\vec{\alpha}})h^{\vec{\alpha}} \sum_{\vec{\alpha}} x_{\vec{\alpha}}(a,b)h^{\vec{\alpha}} \sum_{\vec{\alpha}} S(b_{\vec{\alpha}})h^{\vec{\alpha}}\right) \\
&= -\left(\sum_{\vec{\alpha}}\left(\sum_{\vec{\beta}+\vec{\delta}+\vec{\gamma}=\vec{\alpha}} S\big(a_{\vec{\beta}}\big)x_{\vec{\alpha}}(a,b)S\big(b_{\vec{\gamma}}\big)\right)h^{\vec{\alpha}}\right).
\end{aligned}$$

Comparing like coefficients, we get the desired result.

More generally, suppose that a and b are as above, but we have a factorization $a = ge$ and $b = \tilde{g}\tilde{e}$ in $G(A_{\vec{h}})$, such that $\epsilon(g) = \epsilon(e) = \epsilon(\tilde{g}) = \epsilon(\tilde{e}) = 1$. Let $z, \tilde{z}$ be elements of $A_{\vec{h}}$ such that $\epsilon(z) = \epsilon(\tilde{z}) = 0$, and

$$\Delta_{\vec{h}}(z) = z \otimes g + g \otimes z,$$

$$\Delta_{\vec{h}}(\tilde{z}) = \tilde{z} \otimes \tilde{g} + \tilde{g} \otimes \tilde{z}.$$

Write $ze - \tilde{z}\tilde{e}$ as a power series in $\vec{h}$:

$$ze - \tilde{z}\tilde{e} = \sum_{\vec{\alpha}} z_{\vec{\alpha}}(z, e, \tilde{z}, \tilde{e})\vec{h}^{\vec{\alpha}},$$

where $z_{\vec{\alpha}}(z, e, \tilde{z}, \tilde{e}) \in A$ for every $\tilde{\alpha}$. Let $I(g, \tilde{g}, e, \tilde{e}, z, \tilde{z})$ be the two-sided ideal of A generated by the $x_{\vec{\alpha}}(a, b)$ and the $z\vec{\alpha}(z, e, \tilde{z}, \tilde{e})$, for every $\vec{\alpha}$.

THEOREM 1.0.5. *Suppose that $A_{\vec{h}} = A_0$ is the trivial deformation of A; and $a, b, g, e, z, \tilde{g}, \tilde{e}$, and $\tilde{z}$ are as above. Then $I(g, \tilde{g}, e, \tilde{e}, z, \tilde{z})$ is a Hopf ideal of A.*

Proof. One does calculations like those of the previous theorem based directly on properties (iii) and (iv) of Lemma 1.0.1, noting that $\tilde{g}\tilde{e} = a$ and $\tilde{g}\tilde{e} = b$. For example, to show that I is a coideal in A, we write a, b and $a - b$ as power series exactly as in the previous theorem, and expand $ze - \tilde{z}\tilde{e}$ as above. Then, using properties (iii) of Lemma 1.0.1, we have

$$\Delta_{\vec{h}}(ze - \tilde{z}\tilde{e}) = (ze - \tilde{z}\tilde{e}) \otimes a + a \otimes (ze - \tilde{z}\tilde{e}) \\ + \tilde{z}\tilde{e} \otimes (a - b) + (a - b) \otimes \tilde{z}\tilde{e}.$$

So,

$$\sum_{\vec{\alpha}} \Delta\big(z_{\vec{\alpha}}(z, e, \tilde{z}, \tilde{e})\big)\vec{h}^{\vec{\alpha}} \\ = \sum_{\vec{\alpha}} \sum_{\vec{\beta} + \vec{\gamma} = \vec{\alpha}} \Big(z_{\vec{\gamma}} \otimes a_{\vec{\beta}} + a_{\vec{\gamma}} \otimes z_{\vec{\beta}} + w_{\vec{\gamma}} \otimes x_{\vec{\beta}} + x_{\vec{\gamma}} \otimes w_{\vec{\beta}}\Big)\vec{h}^{\vec{\alpha}},$$

where $w_{\vec{\beta}}$ is the $\vec{\beta}$th coefficient of $\tilde{z}\tilde{e}$. Comparing like coefficients, we see that $\Delta(z_{\vec{\alpha}})$ is an element of $A \otimes I + I \otimes A$, for every $\vec{\alpha}$.

2. EXAMPLES

Let U be the free associative algebra over $\mathbb{Z}$ on a countably infinite set $X = \{X_1, X_2, \ldots, X_n, \ldots\}$; for the convenience, set $X_0 = 1$. Then U is a Hopf algebra over $\mathbb{Z}$ with respect to the diagonal defined by

$$\Delta(X_n) = \sum_{i=0}^{n} X_i \otimes X_{n-i}.$$

The counit ϵ is the map taking a polynomial in the noncommuting variables X_i to its constant term. The values of the antipode S of U may be inductively computed using the formulas

$$\sum_{i=0}^{n} S(X_i)X_{n-i} = \sum_{i=0}^{n} X_i S(X_{n-i}) = 0,$$

which hold for all $n > 0$. For example, $S(X_1) = -X_1$. Notice that since U is cocommutative, $S^2 = \mathrm{id}$, so that S is invertible and is equal to its own inverse.

Let $U_0 = U[[h_1, h_2]]$ be the trivial two-parameter deformation of U. Let $\alpha, \beta, \delta, \gamma$ be any four power series in $\mathbb{Z}[[h_1, h_2]]$ with no constant term. Let $f(t) = \sum_{i=0}^{\infty} X_i t^i$; $f(t)$ is clearly a curve in U_0. Thus, $f(\alpha), f(\beta), f(\delta)$, and $f(\gamma)$ are in $G(U_0)$; this implies that $a = f(\alpha)f(\beta)$ and $b = f(\delta)f(\gamma)$ are also in $G(U_0)$. By the theorems in the previous section, the coefficients of the element

$$f(\alpha)f(\beta) - f(\delta)f(\gamma)$$

in U_0 generate a Hopf ideal $J(\alpha, \beta, \delta, \gamma, f)$ in U, so that $U/J(\alpha, \beta, \delta, \gamma, f)$ is a Hopf algebra over the integers.

There is of course nothing special about having only two variables. In particular, if $U[[z_0, z_1, \dots, z_N]]$ is the trivial $(N+1)$-parameter deformation of U, and α, β, δ, and γ are any four power series in $z_0, z_1, \dots, z_N$ with no constant term, then the coefficients of

$$f(\alpha)f(\beta) - f(\delta)f(\gamma)$$

generate a Hopf ideal of U.

We have deliberately not assigned a grading to the elements X_i so far, but one may do this, as follows: choose an integer K, not equal to 0, and assign $\deg(X_i) = Ki$. Then U becomes a graded Hopf algebra over the integers. We only point out that the multiplication on $U \otimes U$ that we want to use is *not* the graded multiplication

$$(a \otimes b)(c \otimes d) = (-1)^{\deg(b)\deg(c)}(ac \otimes bd)$$

but the standard multiplication

$$(a \otimes b)(c \otimes d) = (ac \otimes bd).$$

(If the reader does not like this, he or she may use only even degrees for the X_i's.)

In any case, if one graded U, then we should impose a further condition on the elements $\alpha, \beta, \delta, \gamma$ above; namely, that they are all homogeneous polynomials of the same degree. In this case, the generators that define $J(\alpha, \beta, \delta, \gamma, f)$ are homogeneous elements of U, and if $J_0(\alpha, \beta, \delta, \gamma, f)$ is the homogeneous two-sided ideal generated by these homogeneous elements, we see that in fact $J_0(\alpha, \beta, \delta, \gamma, f)$ is a graded Hopf ideal in U, and that $U/J_0(\alpha, \beta, \delta, \gamma, f)$ is a graded Hopf algebra over the integers.

EXAMPLE 2.0.6. Let

$$\alpha = h_1, \beta = h_2, \delta = h_2, \gamma = h_1.$$

Then $U/J(\alpha, \beta, \delta, \gamma, f) = \mathbb{Z}[X_1, \dots, X_n, \dots]$, the polynomial ring in the commuting indeterminates $X_1, \dots, X_n, \dots$, with coalgebra map $\Delta(X_n) = \sum_{i=0}^{n} X_i \otimes X_{n-i}$.

Let $\Lambda(\mathbb{Z})$ be the free abelian group on the countably infinite set of generators $T_1, T_2, \dots, T_n, \dots$, grade $\Lambda(\mathbb{Z})$ by setting $\deg(T_i) = i$. Put a ring structure on $\Lambda(\mathbb{Z})$ by the formula

$$T_i T_j = \binom{i+j}{j} T_{i+j}.$$

In other words, $\Lambda(\mathbb{Z})$ is the divided polynomial ring on one generator over the integers. Also, $\Lambda(\mathbb{Z})$ is a Hopf algebra with comultiplication defined by

$$\Delta(T_i) = \sum_{j=0}^{i} T_j \otimes T_{i-j}.$$

EXAMPLE 2.0.7. Suppose that $N \geq 2$ is a fixed integer, and $z_0, z_1, \dots, z_N$ are $N+1$ commuting indeterminates. Assign $\deg(X_i) = i$, and let $\alpha = z_0 + z_1 + \cdots + z_{N-1}$, $\beta = z_N$, $\delta = z_1 + z_2 + \cdots + z_N$, $\gamma = z_0$. Then U/J_0, in the case, is isomorphic as a Hopf algebra to $\Lambda(\mathbb{Z})$.

The proof of this is as follows. Define a surjective homomorphism of Hopf algebras

$$\theta \colon U \to \Lambda(\mathbb{Z})$$

by

$$\theta(X_i) = T_i.$$

The ideal J_0 is generated by the coefficients of the power series

$$\sum_{i,j} X_i X_j (z_0 + \cdots + z_{N-1})^i z_N^j - \sum_{i,j} X_i X_j (z_1 + \cdots + z_N)^i z_0^j,$$

which by the multinomial theorem is equal to

$$\sum_{i,j} X_i X_j \sum_{i_0 + \cdots + i_{N-1} = i} \binom{i}{i_0 \cdots i_{N-1}} z_0^{i_0} \cdots z_{N-1}^{i_{N-1}} z_N^j$$
$$- \sum_{i,j} X_i X_j \sum_{j_1 + \cdots + j_N = i} \binom{i}{j_1 \cdots j_N} z_1^{j_1} \cdots z_N^{j_N} z_0^j.$$

Comparing coefficients of the monomial $z_0^{t_0} z_1^{t_1} \cdots z_N^{t_N}$ for any appropriate $(t_0, \dots, t_N)$, we see that J_0 is generated by the quadratic polynomials

$$X_{t_0 + \cdots + t_{N-1}} X_{t_N} \binom{t_0 + \cdots + T_{N-1}}{t_0 \cdots t_{N-1}} - X_{t_1 + \cdots + t_N} X_{t_0} \binom{t_1 + \cdots + t_N}{t_1 \cdots t_N},$$

for every $(t_0, \ldots, t_N)$. However, one may quickly prove by induction that, in $\Lambda(\mathbb{Z})$,

$$T_{p_1}T_{p_2}\ldots T_{p_r} = \begin{pmatrix} p_1 + \cdots + p_r \\ p_1 \cdots p_r \end{pmatrix} T_{p_1 + \cdots + p_r}.$$

Thus the quadratic polynomials generating J_0 above are all sent to zero via θ. On the other hand, there is a homomorphism of abelian groups from $\Lambda(\mathbb{Z})$ to U/J_0 defined by $T_i \mapsto X_i$. (We use the symbol X_i for both the generator X_i of U and its image in U/J_0.) This homomorphism is a ring homomorphism, since in U/J_0, letting $t_0 = t_2 = \cdots = t_{N-1} = 0$, and $t_1 = i$, $t_N = j$, we see that

$$X_{i+j}\begin{pmatrix} i+j \\ i \end{pmatrix} = X_i X_j.$$

EXAMPLE 2.0.8. Assign $\deg(X_i) = i$, and let

$$\alpha = h_1^2 + h_1 h_2, \qquad \beta = h_2^2, \qquad \delta = h_2^2 + h_1 h_2, \qquad \gamma = h_1^2.$$

Then we set $U/J_0(\alpha, \beta, \delta, \gamma, f) = \mathscr{A}_2(\mathbb{Z})$, the *integral version of the mod*-2 *Steenrod algebra*.

According to Bullett and Macdonald [1], the reduction mod-2 of this algebra over the integers is the mod-2 Steenrod algebra $\mathscr{A}_2$. This algebra was also studied in [9].

EXAMPLE 2.0.9. Assign $\deg(X_i) = 2i(p-1)$, where p is any prime. Let

$$\alpha = h_1^p + h_1^{p-1}h_2 + \cdots + h_1 h_2^{p-1}, \qquad \beta = h_2^p,$$
$$\delta = h_2^p + h_1 h_2^{p-1} + \cdots + h_1^{p-1}h_2, \qquad \gamma = h_1^p.$$

Then we set $U/J_0(\alpha, \beta, \delta, \gamma, f) = \mathscr{A}_p^{ev}(\mathbb{Z})$, the *integral version of the even part of the* mod-*p Steenrod algebra*.

Again, according to Bullett and Macdonald [1], if p is an odd prime, the reduction mod-p of this algebra is the mod-p Steenrod algebra modulo the Bockstein.

EXAMPLE 2.0.10. Fix a positive integer N. Assign $\deg(X_i) = 2i\chi_N$, where χ_N is an integer depending on N. (For example, if $N = p$ is an odd prime, we might let $\chi_p = p - 1$; this would correspond to the traditional grading for the Steenrod algebra.) Let

$$\alpha = h_1^N + h_1^{N-1}h_2 + \cdots + h_1 h_2^{N-1}, \qquad \beta = h_2^N,$$
$$\delta = h_2^N + H_1 h_2^{N-1} + \cdots + h_1^{N-1}h_2, \qquad \gamma = h_1^N.$$

Then, again, $U/J_0(\alpha, \beta, \delta, \gamma, f)$ is a Hopf algebra over the integers, which we call $\mathscr{A}_N^{ev}(\mathbb{Z})$.

In the next example, we take quotients of an extension of U. Let ∂ be a symbol different from, and not commuting with, any of the symbols $X_1, X_2, \ldots$. Let U_∂ be the free associative algebra over the integers on the symbols $\partial, X_1, \ldots, X_n, \ldots$ modulo the two-sided ideal generated by ∂^2. Grade U_∂ by setting $\deg(X_i) = 2i\chi_N$ as in the previous example and $\deg(\partial) = 1$. Now, on $U_\partial \otimes U_\partial$ we use the graded multiplication! (Note that this means we use this multiplication on $U_\partial \otimes U_\partial$ throughout all the definitions of algebra and Hopf algebra.)

We may then make U_∂ into a Hopf algebra by setting $\Delta(\partial) = \partial \otimes 1 + 1 \otimes \partial$. (The graded multiplication on $U_\partial \otimes U_\partial$ ensures that the two-sided ideal generated by ∂^2 in the free associative algebra is a coideal, so that this definition makes sense. The symbol ∂ corresponds to the Bockstein operator in the Steenrod algebra.)

Let $U_{\partial,0}$ be the trivial two-parameter deformation of U_∂ with parameters h_1, h_2 as above. Suppose $\alpha, \beta, \delta, \gamma$ are as in the previous example; let $g = f(\alpha)$, $e = f(\beta)$, $\tilde{g} = f(\delta)$, $\tilde{e} = f(\gamma)$, set $a = ge$ and $b = \tilde{g}\tilde{e}$. Finally, set $z = h_2(\partial g - g\partial)$ and $\tilde{z} = h_1(\partial\tilde{g} - \tilde{g}\partial)$. Naturally we consider all of these as elements of $U_{\partial,0}$. One may verify:

(i) Using Lemma 1.0.2, $g, e, \tilde{g}, \tilde{e}, z, \tilde{z}$ all satisfy the properties of the elements with the same names in Lemma 1.0.1, even with the graded multiplication on $U_\partial \otimes U_\partial$! This is because the coefficients of all the grouplike elements are of even degree.

(ii) Because the coefficients of all the grouplike elements have even degree, the conclusions of Lemma 1.0.1 remain true as well. For example, performing a piece of the computation proving part (iii) of Lemma 1.0.1, if $g = \sum g_{\vec\alpha}\vec{h}^{\vec\alpha}$, $e = \sum e_{\vec\alpha}\vec{h}^{\vec\alpha}$, and $z = \sum z_{\vec\alpha}\vec{h}^{\vec\alpha}$, where we assume that all of the coefficients of g and e have even degree, then in $(U_\partial \otimes U_\partial)[[\vec{h}]]$,

$$
\begin{aligned}
(z \otimes g)(e \otimes e) &= \Big(\sum z_{\vec\alpha}\vec{h}^{\vec\alpha} \otimes \sum g_{\vec\alpha}\vec{h}^{\vec\alpha}\Big)\Big(\sum e_{\vec\alpha}\vec{h}^{\vec\alpha} \otimes \sum e_{\vec\alpha}\vec{h}^{\vec\alpha}\Big) \\
&= \sum_{\vec\alpha}\sum_{\vec\alpha}\sum_{\vec\delta}\sum_{\vec\gamma}\big(z_{\vec\alpha} \otimes g_{\vec\beta}\big)\big(e_{\vec\delta} \otimes e_{\vec\gamma}\big)\vec{h}^{\vec\alpha+\vec\beta+\vec\delta+\vec\gamma} \\
&= \sum_{\vec\alpha}\sum_{\vec\beta}\sum_{\vec\delta}\sum_{\vec\gamma}\big(z_{\vec\alpha}e_{\vec\delta} \otimes g_{\vec\beta}e_{\vec\gamma}\big)\vec{h}^{\vec\alpha+\vec\beta+\vec\delta+\vec\gamma},
\end{aligned}
$$

since the coefficients of e have even degree. In other words,

$$(z \otimes g)(e \otimes e) = ze \otimes ge.$$

Thus we may use Theorem 1.0.5 to conclude that the ideal of Theorem 1.0.5 is in this particular case a Hopf ideal.

This allows us to exhibit

EXAMPLE 2.0.11. Let $\mathscr{A}_N(\mathbb{Z})$ be the quotient of the Hopf algebra U_∂ (with graded multiplication on $U_\partial \otimes U_\partial$) by the two-sided ideal generated by the coefficients of the two power series

$$f\left(h_1^N + h_1^{N-1}h_2 + \cdots + h_1 h_2^{N-1}\right) f\left(h_2^N\right)$$
$$- f\left(h_2^N + h_1 h_2^{N-1} + \cdots + h_1^{N-1}h_2\right) f\left(h_1^N\right)$$

and

$$h_2\Big(\partial f\left(h_1^N + h_1^{N-1}h_2 + \cdots + h_1 h_2^{N-1}\right)$$
$$- f\left(h_1^N + h_1^{N-1}h_2 + \cdots + h_1 h_2^{N-1}\right)\partial \Big) f\left(h_2^N\right)$$
$$- h_1\Big(\partial f\left(h_2^N + h_1 h_2^{N-1} + \cdots + h_1^{N-1}h_2\right)$$
$$- f\left(h_2^N + h_1 h_2^{N-1} + \cdots + h_1^{N-1}h_2\right)\partial \Big) f\left(h_1^N\right).$$

Then $\mathscr{A}_N(\mathbb{Z})$ is a Hopf algebra.

If $N = p$ is an odd prime, then the mod-p reduction of $\mathscr{A}_N(\mathbb{Z})$ is a the mod-p Steenrod algebra, by the work of Bullett and Macdonald [1].

In fact, there is another "generalization" of the Steenrod algebras, which one may construct as follows:

EXAMPLE 2.0.12. Let $A_N(\mathbb{Z})$ be the quotient of the Hopf algebra U_∂ (with graded multiplication on $U_\partial \otimes U_\partial$) by the two-sided ideal generated by the coefficients of the two power series

$$f\left(h_1(h_1 - h_2)^{N-1}\right) f\left(h_2^N\right) - f\left(h_2(h_1 - h_2)^{N-1}\right) f\left(h_1^N\right)$$

and

$$h_2\Big(\partial f\left(h_1(h_1 - h_2)^{N-1}\right) - f\left(h_1(h_1 - h_2)^{N-1}\right)\partial \Big) f\left(h_2^N\right)$$
$$- h_1\Big(\partial f\left(h_2(h_1 - h_2)^{N-1}\right) - f\left(h_2(h_1 - h_2)^{N-1}\right)\partial \Big) f\left(h_1^N\right).$$

Then $A_N(\mathbb{Z})$ is a Hopf algebra.

Again, if $N = p$ is an odd prime, then the mod-p reduction of $A_N(\mathbb{Z})$ is the mod-p Steenrod algebra by [1].

If k is any commutative ring with 1, then $U(k) = U \otimes_{\mathbb{Z}} k$ (the free associative algebra on the set $\{X_1, \ldots, X_n, \ldots\}$ over k) is a Hopf algebra over k, with the same diagonal map as above, and we may define Hopf algebras $\mathscr{A}_N^{ev}(k)$ following the procedure in the above examples.

EXAMPLE 2.0.13. Suppose that k is a commutative ring. Suppose that q is an element of k. Assign degree $X_i = i$ in $U(k)$. Let

$$\alpha = qh_1, \qquad \beta = h_2, \qquad \delta = h_2, \qquad \gamma = h_1.$$

Then $U(k)/J_0(\alpha, \beta, \delta, \gamma, f)$ is a Hopf algebra with coalgebra map $\Delta(X_n) = \sum_{i=0}^n X_i \otimes X_{n-i}$.

In this example, the coefficient of $h_1^i h_2^j$ in $f(\alpha)f(\beta) - f(\delta)f(\gamma)$ is $q^i X_i X_j - X_j X_i$. Thus, in the quotient algebra (using the symbol X_j for the generator of $U(k)$ and its image is the quotient algebra),

$$q^j X_j = X_j,$$

for all j. Depending on what sort of ring k is, and what sort of element q is, we get various sorts of Hopf algebras here. For example, if k is a field, $q \neq 1$ and $q^i \neq 1$ for all $i > 0$, then in the quotient, $X_j = 0$ for all j; this quotient is the trivial k-algebra k.

On the other hand, if k is a field and q is a primitive rth root of 1 in k, then the only generators remaining in the quotient are $X_r, X_{2r}, \dots, X_{nr}, \dots$; also the equation

$$X_{mr} X_{nr} = q^{mr} X_{mr} X_{nr} = X_{nr} X_{mr}$$

which holds in the quotient, for every m and n, shows that the quotient is the polynomial ring $k[X_r, X_{2r}, \dots]$ on the commuting indeterminates $X_r, X_{2r,\dots}$.

However, if, for example, $k = \mathbb{Z}$ and $q = 2$, the quotient algebra is commutative, with nonzero elements in every degree; also, every positive degree element in the quotient is torsion.

EXAMPLE 2.0.14. Suppose that k is a commutative ring, and that q is an element of k. Assign $\deg(X_i) = i$ in $U(k)$, and let

$$\alpha = h_1^2 + qh_1h_2, \qquad \beta = h_2^2, \qquad \delta = h_2^2 + qh_1h_2, \qquad \gamma = h_1^2.$$

Then $U(k)/J_0(\alpha, \beta, \delta, \gamma, f)$ is a Hopf algebra (with coalgebra map $\Delta(X_n) = \sum_{i=0}^n X_i \otimes X_{n-i}$).

As the reader can see, there are many possibilities.

We return to give just a few more somewhat different types of examples than the above and finally, at the end of this section, we give an initial discussion of example 2.0.10.

The special thing about U and the curve $f(t)$ in the above lies in the fact that U represents "curves over 1," $f(t)$ being the prototypical curve over 1. For a discussion of curves, why U "represents" curves and the importance of curves in the theory of formal groups, we refer the reader to [7]; this is a big book and a big topic, however! In this paper we will not address the connection between Hopf algebras and formal groups.

More generally, in fact, if A is any Hopf algebra over k, and $\phi(t) = \sum_{i=0}^{\infty} a_i t^i$ is any curve in A, then we get a Hopf ideal $J(\alpha, \beta, \delta, \gamma, \phi)$ in A, exactly as above: by taking the coefficients of the element

$$\phi(\alpha)\phi(\beta) - \phi(\delta)\phi(\gamma)$$

in $A[[h_1, h_2]]$ (regarded as the trivial deformation of A) as generators. The power series α, β, δ, and γ are, as before, in $k[[h_1, h_2]]$ and have no constant term. Notice that in order to use Theorem 1.0.5, ϕ must be a curve (which is the same as a grouplike element).

Here is another specific example. We use the definitions and theorems of [7]. Let p be a prime number, and let $U_{(p)} = U(\mathbb{Z}_{(p)})$. Grade $U_{(p)}$ by setting degree $X_i = 2i(p-1)$. If $\zeta = \sum_{i=0}^{\infty} \zeta_i t^i$ is any curve over 1 in $U_{(p)}$, such that degree $\zeta_i = 2i(p-1)$ for every i, there exists a unique homomorphism of graded Hopf algebras

$$\hat{\zeta}: U_{(p)} \to U_{(p)}$$

such that $\hat{\zeta}(X_i) = \zeta_i$ for every i. Thus the image of $\hat{\zeta}$ is a graded Hopf subalgebra of $U_{(p)}$ which we shall call $U_{(p),\zeta}$. Notice that ζ is also a curve in $U_{(p),\zeta}$.

According to [7] (the basic references in [7] for these results are two papers by Ditters [3, 4]; however these papers contain some gaps which have been corrected and improved by Scholtens in her thesis [10]), there exists a *p-pure curve* in the Hopf algebra $U_{(p)}$: this is a curve $E(t) = 1 + \sum_{i=1}^{\infty} E_i t^i$ over 1 in $U_{(p)}$ such that

(a) degree$(E_i) = 2i(p-1)$, for every i,

(b) $U_{(p),E}$ is the subalgebra of $U_{(p)}$ generated by the elements $E_1 = X_1, E_p, E_{p^2}, \ldots, E_{p^n}, \ldots$.

Now we get a lot more Hopf algebras!

EXAMPLE 2.0.15. Let $A = U_{(p),E}$, where E is a p-pure curve as above. Let the polynomials $\alpha, \beta, \delta, \gamma$ be as in, say, Example 2.0.9 above. Then the coefficients of

$$E(\alpha)E(\beta) - E(\delta)E(\gamma)$$

generate a Hopf ideal of $U_{(p), E}$, and we have a new Hopf algebra over $\mathbb{Z}_{(p)}$ by taking the quotient.

Even though we are particularly interested in these examples, these Hopf algebras seem quite complicated, and a discussion of them will be deferred to future work.

Notice that the Hopf ideals in all the examples above are "quadratic" ideals in some sense. We may get "higher-order" ideals by replacing the element

$$\phi(\alpha)\phi(\beta) - \phi(\delta)\phi(\gamma)$$

by elements

$$\phi_1(\alpha_1)\phi_2(\alpha_2)\cdots\,\phi_m(\alpha_m) - \tilde{\phi}_1(\delta_1)\tilde{\phi}_2(\delta_2)\cdots\,\tilde{\phi}_m(\delta_m).$$

Here, the ϕ_is and the $\tilde{\phi}_i$s must be curves, and the α_is and δ_is power series over k with no constant term. Again, the ϕ_is must be curves (grouplike elements) in order to use Theorem 1.0.5. One may build even more general Hopf ideals as well.

The rest of this section is devoted to a discussion of Example 2.0.10. Discussions of some other examples will be deferred to a later paper [6].

The "generating function" for the graded Hopf ideal J_0 in U (by this we mean that the coefficients of the paper series generate the ideal in question) is the power series in U_0

$$\sum_{i=1}^{\infty} X_i\left(h_1^N + h_1^{N-1}h_2 + \cdots + h_1h_2^{N-1}\right)^i \sum_{i=0}^{\infty} X_i h_2^{Ni}$$
$$- \sum_{i=0}^{\infty} X_i\left(h_2^N + h_1h_2^{N-1} + \cdots + h_1^{N-1}h_2\right)^i \sum_{i=0}^{\infty} X_i h_1^{Ni}.$$

First, make a change of variables: set $h_1 = x$ and $h_2 = xy$. (In other words, we are "dehomogenizing.") Thus the above ideal generated by the coefficients of the polynomials

$$\sum_{i=0}^{k} X_i X_{k-i}\left(1 + y + \cdots + y^{N-1}\right)^i y^{N(k-i)}$$
$$- \sum_{i=0}^{k} X_i X_{k-i}\left(1 + y + \cdots + y^{N-1}\right)^i y^i,$$

for all $k \geq 0$.

It will be convenient for our calculations to expand the powers of $1 + y + \cdots + y^{N-1}$ in the following way. If $b \geq 0$ is an integer, we may write

$$\frac{1}{(1-y)^b} = \sum_{r=0}^{\infty} h_r y^r,$$

where $h_r = \binom{b+r-1}{r}$, for every $r \geq 0$.

Also, if $a \geq 0$ is an integer,

$$(1 - y^{a+1})^b = \sum_{s \geq 0} \binom{b}{s} (-1)^s y^{(a+1)s} = \sum_{u \geq 0} \xi_u y^u,$$

where

$$\xi_u = \begin{cases} 0, & (a+1) \nmid u, \\ (-1)^{u/(a+1)} \begin{pmatrix} b \\ \dfrac{u}{a+1} \end{pmatrix}, & (a+1) | u. \end{cases}$$

Thus, the coefficient of y^v in $((1-y^{a+1})/(1-y))^b$ is

$$\xi_v(a,b) = \sum_{\alpha \in S(a,v)} (-1)^{\alpha/(a+1)} \begin{pmatrix} b \\ \dfrac{\alpha}{a+1} \end{pmatrix} \binom{b+v-\alpha-1}{v-\alpha},$$

where

$$S(a,v) = \{\alpha \in \mathbb{Z} | 0 \leq \alpha \leq v, (a+1) | \alpha\}.$$

Notice:

a. $\xi_0(a,b) = 1$, for every a, b.

b. $\xi_v(a,0) = \xi_v(0,b) = \begin{cases} 1, & v = 0, \\ 0, & v > 0, \end{cases}$ for every a, b.

c. $\xi_v(1,b) = \binom{b}{v}$, for all v, b.

d. $\xi_v(a,1) = \begin{cases} 1, & v \leq a, \\ 0, & v > a, \end{cases}$ for every a.

e. $\xi_v(a,b) = 0$ if $v > ab$; $\xi_{ab}(a,b) = 1$.

f. $\xi_v(a,b) = \binom{b+v-1}{v}$, if $v < a+1$.

Now, write

$$(1 + y + \cdots y^{N-1})^i = \sum_{v=0}^{\infty} \xi_v(N-1, i) y^v.$$

Thus the polynomials above may be rewritten as

$$\sum_{i\geq 0}\sum_{v\geq 0} X_i X_{k-i}\,\xi_v(N-1,i)\,y^{v+N(k-i)} - \sum_{i\geq 0}\sum_{v\geq 0} X_i X_{k-i}\,\xi_v(N-1,i)\,y^{i+v}.$$

Let $T \geq 0$, and write down the coefficient of y^T in the above, yielding that the generators of the ideal J_0 are

$$\sum_{i\geq 0} X_i X_{k-i}\,\xi_{T-N(k-i)}(N-1,i) - \sum_{i\geq 0} X_i X_{k-i}\,\xi_{T-i}(N-1,i), \tag{2.0.16}$$

for every $k \geq 0$ and every $T \geq 0$.

Now, let us pass to the quotient $\mathscr{A}_N^{ev}(\mathbb{Z})$ of U; we denote the image of X_i in the quotient by the same symbol X_i.

If $T < N$, then $\xi_{T-Nq}(N-1,b) = 0$, unless $q = 0$. So, for every k, and every T such that $0 \leq T \leq N-1$, we have

$$X_k\,\xi_T(N-1,k) = \sum_{i\geq 0} X_i X_{k-i}\,\xi_{T-i}(N-1,i).$$

But, also, $T < N$ implies that $T - i \leq N$ if $i \geq 0$; thus we have (using facts about the ξ_vs noted above):

$$X_k\binom{k+T-1}{T} = \sum_{i\geq 0} X_i X_{k-i}\binom{T-1}{T-i},$$

for every $k \geq 0$, and every T such that $0 \leq T \leq N-1$.

This gives immediately the following lemma:

LEMMA 2.0.17. *For every $i \geq 0$, $X_1 X_i = (i+1)X_{i+1}$ in $\mathscr{A}_N^{ev}(\mathbb{Z})$.*

Proof. Note that $N \geq 2$, so $1 \leq N-1$. The desired result follows from writing down the generators in (2.0.16) in the case $T = 1$.

In fact, more generally, we have:

LEMMA 2.0.18. *If $k < N$, then for every i such that $0 \leq i \leq k$,*

$$X_i X_{k-i} = \binom{k}{i} X_k$$

in $\mathscr{A}_N^{ev}(\mathbb{Z})$.

Proof. Let $1 \leq i < k$. Suppose that, for every α such that $0 \leq \alpha \leq i$,

$$X_\alpha X_{k-\alpha} = \binom{k}{\alpha} X_k.$$

Then

$$\begin{aligned} X_k \binom{k+i}{1+i} &= \sum_{\alpha=0}^{i} X_\alpha X_{k-\alpha} \binom{i}{i+1-\alpha} + X_{i+1} X_{k-(i+1)} \\ &= \left[\sum_{\alpha=0}^{i} \binom{k}{\alpha} \binom{i}{i+1-\alpha} \right] X_k + X_{i+1} X_{k-(i+1)} \\ &= \left[\binom{k+i}{i+1} - \binom{k}{i+1} \right] X_k + X_{i+1} X_{k-(i+1)}. \end{aligned}$$

Thus,

$$X_k \binom{k}{i+1} = X_{i+1} X_{k-(i+1)}.$$

We also have the following theorem:

THEOREM 2.0.19. *For every $N \geq 2$, there is a surjective homomorphism of Hopf algebras*

$$\theta \colon \mathscr{A}_N^{ev}(\mathbb{Z}) \to \Lambda(\mathbb{Z})$$

such that $\theta(X_i) = T_i$ for every i.

Proof. This is a direct consequence of the following simple fact: there is a ring homomorphism

$$U[[z_0, z_1, \ldots, z_N]] \to U[[h_1, h_2]]$$

defined by

$$z_i \mapsto h_1^{N-i} h_2^i.$$

Since we may define $\Lambda(\mathbb{Z})$ by the coefficients of the power series

$$f(z_0 + \cdots + z_{N-1}) f(z_N) - f(z_1 + \cdots + z_N) f(z_0)$$

and $\mathscr{A}_N^{ev}(\mathbb{Z})$ is defined by the coefficients of the power series

$$f\left(h_1^N + \cdots + h_1 h_2^{N-1}\right) f\left(h_2^N\right) - f\left(h_1^{N-1} h_2 + \cdots + h_2^N\right) f\left(h_1^N\right),$$

the desired result follows.

Thus, in a certain sense, "as $N \to \infty$, $\mathscr{A}_N(\mathbb{Z}) \to \Lambda(\mathbb{Z})$."

Notice that we would have to regrade $\Lambda(\mathbb{Z})$ for every N in order to get a homomorphism respecting the grading in the above theorem.

For the rest of the section, we specialize even further to the case $N = 2$. Let $C = [\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$ be the commutator ideal of $\mathscr{A}_2(\mathbb{Z})$. This is a homogeneous two-sided deal in $\mathscr{A}_2(\mathbb{Z})$.

LEMMA 2.0.20. *For every $i \geq 0$ and $j \geq 0$, in $\mathscr{A}_2(\mathbb{Z})$ we have*

$$X_i X_j \equiv \binom{i+j}{j} X_{i+j} \mod C.$$

Proof. We prove by induction on i: For all $j \geq 0$,

$$X_i X_j \equiv \binom{i+j}{j} X_{i+j} \mod C.$$

For $i = 1$, this is implied by Lemma 2.0.17, in the case $N = 2$.

Now, consider $X_n X_j$, where $n > 1$, and we know that, for all $i < n$ and for all $b \geq 0$,

$$X_i X_b \equiv \binom{i+b}{b} X_{i+b} \mod C.$$

In the basic formulation (2.0.16) for the generators of the ideal defining $\mathscr{A}_2(\mathbb{Z})$, let $T = n$ and $k = n + j$ (and, of course, $N = 2$). So, we see that in $\mathscr{A}_2(\mathbb{Z})$,

$$X_n X_j = \binom{n+j}{n} X_{n+j} + \sum_{i=0}^{n+j-1} \binom{i}{2i-2j-n} X_i X_{n+j-i} - \sum_{i \geq [n/2]}^{n-1} \binom{i}{n-i} X_i X_{n+j-i}.$$

Now, in the second summand on the right-hand side above, change the summation index to k, by setting $k = i - j$:

$$X_n X_j = \binom{n+j}{n} X_{n+j} + \sum_{k \geq [n/2]}^{n-1} \binom{j+k}{2k-n} X_{j+k} X_{n-k} - \sum_{i \geq [n/2]}^{n-1} \binom{i}{n-i} X_i X_{n+j-i}.$$

Passing to the quotient modulo C and using induction, we see that

$$X_n X_j \equiv \binom{n+j}{n} X_{n+j} + \sum_{k \geq [n/2]}^{n-1} \binom{j+k}{2k-n} \binom{n+j}{j+k} X_{j+n}$$
$$- \sum_{i \geq [n/2]}^{n-1} \binom{i}{n-i} \binom{n+j}{i} X_{n+j},$$

modulo C. But, one may check that, for every k in the range indicated above,

$$\binom{j+k}{2k-n} \binom{n+j}{j+k} = \binom{k}{n-k} \binom{n+j}{k}.$$

The following corollary shows that $\mathscr{A}_2(\mathbb{Z})$ made commutative is $\Lambda(\mathbb{Z})$.

COROLLARY 2.0.21. *$\mathscr{A}_2(\mathbb{Z})/[\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$ is isomorphic to $\Lambda(\mathbb{Z})$ as a graded $\mathbb{Z}$-algebra.*

Proof. The surjective ring homomorphism $\theta\colon \mathscr{A}_2(\mathbb{Z}) \to \Lambda(\mathbb{Z})$ was defined in Theorem 2.0.19; since $\Lambda(\mathbb{Z})$ is a commutative ring, we immediately get an induced homomorphism

$$\Theta\colon \mathscr{A}_2(\mathbb{Z})/[\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})] \to \Lambda(\mathbb{Z}).$$

Since $\Lambda(\mathbb{Z})$ is free as an abelian group on the set $\{T_1, \ldots, T_n \ldots\}$, we may define an inverse homomorphism (of abelian groups) to Θ by the formula

$$\Theta^{-1}(T_i) = X_i + [\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$$

in $\mathscr{A}_2(\mathbb{Z})/[\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$. By the above theorem, this function is a homomorphism of rings.

COROLLARY 2.0.22. *Let $\theta\colon \mathscr{A}_2(\mathbb{Z}) \to \Lambda(\mathbb{Z})$ be the homomorphism of Theorem* 2.0.19. *Then θ induces an isomorphism of $\mathbb{Q}$-algebras*

$$\theta \otimes 1\colon \mathscr{A}_2(\mathbb{Z}) \otimes \mathbb{Q} \to \Lambda(\mathbb{Z}) \otimes \mathbb{Q}.$$

Since $\Lambda(\mathbb{Z}) \otimes \mathbb{Q}$ is isomorphic as a $\mathbb{Q}$-algebra to the polynomial ring $\mathbb{Q}[T]$ in one variable over $\mathbb{Q}$ via the homomorphism $T_i \otimes 1 \mapsto T^i/i!$, we have

$$\mathscr{A}_2(\mathbb{Z}) \otimes \mathbb{Q} \cong \mathbb{Q}[T]$$

via the map $X_i \otimes 1 \mapsto T^i/i!$.

Proof. The inverse to the map $X_i \otimes 1 \mapsto T^i/i!$ is the ring homomorphism given by $T \mapsto X_1 \otimes 1$: by Lemma 2.0.17 and an easy induction argument,

$$X_1^i = i! X_i$$

in $\mathscr{A}_2(\mathbb{Z})$ for every $i \geq 0$.

COROLLARY 2.0.23. *The set of the torsion elements in $\mathscr{A}_2(\mathbb{Z})$ is equal to the commutator ideal $[\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$.*

Proof. One has the following exact sequence of graded abelian groups (which are finitely generated in each degree):

$$0 \to [\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})] \to \mathscr{A}_2(\mathbb{Z}) \to \Lambda(\mathbb{Z}) \to 0$$

given by the above lemmas. Thus $\mathscr{A}_2(\mathbb{Z})/[\mathscr{A}_2(\mathbb{Z}), A_2(\mathbb{Z})]$ is torsion free.
Tensoring the exact sequence with the rational numbers we get

$$0 \to [\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})] \otimes \mathbb{Q} \to \mathscr{A}_2(\mathbb{Z}) \otimes \mathbb{Q} \to \Lambda(\mathbb{Z}) \otimes \mathbb{Q} \to 0.$$

Again, by previous lemmas, the right-hand arrow here is an isomorphism, forcing the left-hand abelian group to be zero. Thus, $[\mathscr{A}_2(\mathbb{Z}), \mathscr{A}_2(\mathbb{Z})]$ consists entirely of torsion elements.

We point out that lemmas similar to some of the above lemmas are also true for the mod-2 Steenrod algebra [12].
Finally, we would like to point out the following interesting fact. Suppose that instead of the homogeneous two-sided ideal J_0 of U generated by the coefficients of the generating function

$$f(h_1^2 + h_1 h_2) f(h_2^2) - f(h_2^2 + h_1 h_2) f(h_1^2),$$

we take instead the homogeneous ideal I_0 of U generated by the "Adem relations" (for $0 < i < 2j$)

$$X_i X_j = \sum_{k=0}^{[i/2]} \binom{j-1-k}{i-2k} X_{i+j-k} X_k,$$

and form the graded $\mathbb{Z}$-algebra U/I_0. (Here, we interpret the binomial coefficients as the integers that they in fact are, and use the convention that $\binom{n}{m}$ is zero if $m > n$.) Then I_0 is not a Hopf ideal in $U, U/I_0$ is not a Hopf algebra with respect to the induced coalgebra map, and, in fact, U/I_0 cannot be isomorphic to U/J_0 as an algebra. To see this note that the Adem relations above imply that

$$X_1^2 = 0$$

in U/I_0. Thus, again by considering the possible relations among the generators of U/I_0 in degree 1 and 2 implied by the Adem relations above, we see that, in degree 1, U/I_0 is a free abelian group generated by X_1 and, in degree 2, U/I_0 is a free abelian group generated by X_2. Thus, the elements of $U/I_0 \otimes U/I_0$ of degree 2 form a free abelian group generated by $X_1 \otimes X_1$, $1 \otimes X_2$, and $X_2 \otimes 1$. If the coalgebra map Δ of U descended to give a coalgebra map Δ for U/I_0, then on the one hand,

$$\Delta\left(X_1^2\right) = 0,$$

while on the other,

$$\Delta\left(X_1^2\right) = (1 \otimes X_1 + X_1 \otimes 1)^2 = 2(X_1 \otimes X_1)$$

in U/I_0. But, as noted above, $X_1 \otimes X_1$ cannot be a torsion element of $U/I_0 \otimes U/I_0$. This also shows that U/I_0 cannot be isomorphic to U/J_0 as a graded algebra, because X_1 is a nonzero element of degree one in U/I_0 such that $X_1^2 = 0$ in U/I_0, yet no nonzero element of degree 1 in U/J_0 has square equal to zero in U/J_0.

One might say (if sense could be made of it) that $\mathscr{A}_2(\mathbb{Z})$ represents a group scheme over the integers whose reduction mod 2 is represented by the mod-2 Steenrod algebra. (Unfortunately, $\mathscr{A}_2(\mathbb{Z})$ is neither finitely generated as an algebra over the integers, nor is it commutative, nor is it torsion free; thus this description is not to be taken literally.)

3. DEFORMATIONS BY TWISTING

According to Drinfel'd [5] (see also [8]), a (topological) *quasi-bialgebra* Q over a commutative (topological) ring R is an associative (topological) R-algebra with 1, with homomorphisms of algebras $\Delta\colon Q \to Q \hat{\otimes} Q$ (the comultiplication) and $\epsilon\colon Q \to R$ (the counit), such that:

(a) There exists an invertible element $\Phi \in Q \hat{\otimes} Q \hat{\otimes} Q$ (the *Drinfel'd associator* for Q) with

$$(\mathrm{id} \otimes \Delta)(\Delta(a)) = \Phi((\Delta \otimes \mathrm{id})(\Delta(a)))\Phi^{-1}$$

and

$$(\mathrm{id} \otimes \mathrm{id} \otimes \Delta)(\Phi)(\Delta \otimes \mathrm{id} \otimes \mathrm{id})(\Phi) = (1 \otimes \Phi)(\mathrm{id} \otimes \Delta \otimes \mathrm{id})(\Phi)(\Phi \otimes 1),$$

for all $a \in Q$.

(b) There exist invertible elements $l, r \in Q$ such that, for all $a \in Q$,

$$(\epsilon \otimes \mathrm{id})(\Delta(a)) = l^{-1}al,$$
$$(\mathrm{id} \otimes \epsilon)(\Delta(a)) = r^{-1}ar.$$

and

$$(\mathrm{id} \otimes \epsilon \otimes \mathrm{id})(\Phi) = r \times l^{-1}.$$

The elements l, r in the above definition are called the *unit constraints* for Q.

From now on, we delete the word "topological."

Drinfel'd [5, 8] calls a quasi-bialgebra Q (notation as above) a *quasi-Hopf algebra* if there exist an invertible antiautomorphism S of Q and elements α and β of Q (which we shall call the "antipode constraints" for Q) such that if $a \in Q$ and $\Delta(a) = \sum a' \otimes a''$, then

(a) $\sum S(a')\alpha a'' = \epsilon(a)\alpha$, $\sum a'\beta S(a'') = \epsilon(a)\beta$

and

(b) $\sum_i \Phi_{1i}\, \beta S(\Phi_{2i})\alpha \Phi_{3i} = r^{-1}l$, $\sum_i S((\Phi^{-1})_{1i})\alpha(\Phi^{-1})_{2i}\, \beta S((\Phi^{-1})_{3i}) = S(r)S(l)^{-1}$, where $\Phi = \sum_i \Phi_{1i} \otimes \Phi_{2i} \otimes \Phi_{3i}$ and $\Phi^{-1} = \sum_i (\Phi^{-1})_{1i} \otimes (\Phi^{-1})_{2i} \otimes (\Phi^{-1})_{3i}$.

The map S will be called the antipode for Q. Actually, Drinfel'd makes the definition of a quasi-Hopf algebra explicitly only in the case $r = l = 1$, but in fact the definition above is implicitly contained in his work.

Drinfel'd [5] comments that "In the definition of quasi-Hopf algebra, before the words "there exist" should probably be inserted "locally with respect to Spec(k)." [We wrote R instead of k in the above.] We are not doing this, because we are primarily concerned with the case that k is a field or the ring of formal series over a field." Since we are not necessarily interested in working over fields, perhaps Drinfel'd's suggested insertion should be made.

We broaden our definition of "deformation" of a Hopf algebra A to also allow deformations $A_{\vec{h}}$ that are perhaps only quasi-Hopf algebras. We will refer to such a deformation as a "quasi-Hopf algebra deformation."

Here, we discuss various examples of quasi-Hopf algebra of deformations obtained by "twisting" (or "skrooching" [11]).

Again, we work over the integers and restrict our attention to U as above. Consider the trivial one-parameter deformation $U[[h]]$ of U. Now, fix any two curves $E(t) = \sum_{i=0}^{\infty} E_i t^i$ and $F(t) = \sum_{i=0}^{\infty} F_i t^i$ over 1 in the power series ring $U[[t]]$ such that $\epsilon(E_i) = \epsilon(F_i) = 0$ for all $i > 0$. (If one is interested in working in a graded context, as we will be later in [6], one may make various homogeneity conditions on the E_is and F_is.)

Let $\mathscr{F}_{E,F} = E(h) \otimes F(h)$ in the ring $U[[h]] \hat{\otimes}_{\mathbb{Z}[[h]]} U[[h]]$. Let $SE(t) = \sum_{i=0}^{\infty} S(E_i)t^i$ and $SF(t) = \sum_{i=0}^{\infty} S(F_i)t^i$; note that $E(h)SE(h) = SE(h)E(h) = F(h)SF(h) = SF(h)F(h) = 1$, so that $\mathscr{F}_{E,F}^{-1} = SE(h) \otimes SF(h)$. Define a new homomorphism of $\mathbb{Z}$-algebras

$$\Delta_{E,F}: U[[h]] \to U[[h]] \hat{\otimes}_{\mathbb{Z}[[h]]} U[[h]]$$

by the formula

$$\begin{aligned} \Delta_{E,F}(X_i) &= (E(h) \otimes F(h))\Delta(X_i)(SE(h) \otimes SF(h)) \\ &= \mathscr{F}_{E,F}\Delta(X_i)\mathscr{F}_{E,F}^{-1}. \end{aligned}$$

THEOREM 3.0.24. *$\Delta_{E,F}$ defines a quasi-Hopf algebra deformation of U, which we will call $U^{E,F}$. The multiplication, antipode, unit, and counit of $U^{E,F}$ remain the same as in the trivial deformation (thus are obtained by extension from U).*

The Drinfel'd associator Φ for $U^{E,F}$ is $\Phi = SE(h) \otimes E(h)F(h)SE(h)SF(h) \otimes F(h)$. The elements l, r required in the definition are $l = SF(h)$ and $r = SE(h)$. The elements α and β are $\alpha = \beta = r^{-1}l = E(h)SF(h)$.

Proof. In fact, the proof of this theorem is a straightforward enough calculation, by now essentially standard in the quantum group setting, and usually left as an exercise to the reader (in, for example, [2], [8], [5], and [11]). However, we write down some details here because it interests us to do so, and also because we are not in the case $r = l = 1$. Also, the calculations do not require U to be a free algebra, but only that U is a Hopf algebra with respect to Δ, ϵ, etcetera. So, in fact, we may replace U by any Hopf algebra, and E, F by any curves in the Hopf algebra.

Since U is a free algebra on the X_ns we need only check the conditions on each X_n. Now, we first point out that

$$\Delta_{E,F}(\xi) = (E(h) \otimes F(h))\Delta(\xi)(SE(h) \otimes SF(h))$$

for every element ξ in U; thus

$$\Delta_{E,F}\left(\sum_{n=0}^{\infty} \xi_n h^n\right) = (E(h) \otimes F(h)) \sum_{n=0}^{\infty} (\Delta(\xi_n)h^n)(SE(h) \otimes SF(h)),$$

for every element $\sum_{n=0}^{\infty} \xi_n h^n$ of $U[[h]]$. Using this, we see that

$$\begin{aligned} \Delta_{E,F}(E(h)) &= (E(h) \otimes F(h))(E(h) \otimes E(h))(SE(h) \otimes SF(h)) \\ &= E(h) \otimes F(h)E(h)SF(h), \\ \Delta_{E,F}(SE(h)) &= SE(h) \otimes F(h)SE(h)SF(h), \\ \Delta_{E,F}(F(h)) &= E(h)F(h)SE(h) \otimes F(h), \\ \Delta_{E,F}(SF(h)) &= E(h)SF(h)SE(h) \otimes SF(h). \end{aligned}$$

Thus,

$$\Delta_{E,F}(X_n) = \sum_{i=0}^{n} E(h) X_i SE(h) \otimes F(h) X_{n-i} SF(h),$$

and so

$$\begin{aligned}(1 \otimes \Delta_{E,F})(\Delta_{E,F}(X_n)) \\ &= \sum_{l=0}^{n} \sum_{k=0}^{i} E(h) X_k SE(h) \otimes E(h) F(h) X_{l-k} SF(h) SE(h) \\ &\quad \otimes F(h)^2 X_{n-l} SF(h)^2.\end{aligned}$$

On the other hand, a similar computation yields

$$\begin{aligned}(\Delta_{E,F} \otimes 1)(\Delta_{E,F}(X_n)) \\ &= \sum_{l=0}^{n} \sum_{k=0}^{i} E(h)^2 X_k SE(h)^2 \otimes F(h) E(h) X_{l-k} SE(h) SF(h) \\ &\quad \otimes F(h) X_{n-l} SF(h).\end{aligned}$$

Conjugating by Φ above, we get

$$(\Phi)(\Delta_{E,F} \otimes 1)(\Delta_{E,F}(X_n))(\Phi)^{-1} = (1 \otimes \Delta_{E,F})(\Delta_{E,F}(X_n)).$$

Also (from now on, we write $E(h) = E$, $F(h) = F$, $SE(h) = E^{-1}$, and $SF(h) = F^{-1}$),

$$\begin{aligned}&(E^{-1} \otimes EFE^{-1}F^{-1} \otimes EFE^{-1} \otimes F)(E^{-1} \otimes FE^{-1}F^{-1} \otimes EFE^{-1}F^{-1} \otimes F) \\ &\quad = E^{-2} \otimes EFE^{-2}F^{-1} \otimes EF^2E^{-1}F^{-1} \otimes F^2 \\ &\quad = (1 \otimes E^{-1} \otimes EFE^{-1}F^{-1} \otimes F) \\ &\qquad \times (E^{-1} \otimes E^2FE^{-1}F^{-1}E^{-1} \otimes FEFE^{-1}F^{-2} \otimes F) \\ &\qquad \times (E^{-1} \otimes EFE^{-1}F^{-1} \otimes F \otimes 1).\end{aligned}$$

Checking the condition for l and r, we have (for all n)

$$(\epsilon \otimes \mathrm{id})\Delta_{E,F}(X_n) = \sum_{i=0}^{n} \epsilon(EX_iE^{-1}) \otimes FX_{n-i}F^{-1} = FX_nF^{-1} = l^{-1}X_n l,$$

similarly

$$(\mathrm{id} \otimes \epsilon)\Delta_{E,F}(X_n) = EX_nE^{-1} = r^{-1}X_n r.$$

Also,

$$(\mathrm{id} \otimes \epsilon \otimes \mathrm{id})(E^{-1} \otimes EFE^{-1}F^{-1} \otimes F) = E^{-1} \otimes F = r \otimes l^{-1}.$$

For the assertion about the antipode, for all $n > 0$,

$$\begin{aligned}\sum_{i=0}^{n} EX_i E^{-1}EF^{-1}S\left(FX_{n-i}F^{-1}\right) &= \sum_{i=0}^{n} EX_i E^{-1}EF^{-1}FS(X_{n-i})F^{-1} \\ &= \sum_{i=0}^{n} EX_i S(X_{n-i})F^{-1} = \epsilon(X_n)EF^{-1},\end{aligned}$$

and similarly

$$\sum_{i=0}^{n} ES(X_i)E^{-1}EF^{-1}FX_{n-i}F^{-1} = \epsilon(X_n)DF^{-1},$$

Finally, $E^{-1}EF^{-1}FEF^{-1}E^{-1}EF^{-1}F = EF^{-1} = \alpha$ so that indeed S furnishes the antipode for $\Delta_{E,F}$.

We also point out that $U^{E,F}$ is "almost cocommutative" with "universal $\mathscr{R}$-matrix" equal to $FD^{-1} \otimes EF^{-1}$. In fact, one can quickly check that $\mathscr{R} = FE^{-1} \otimes EF^{-1}$ satisfies the "Yang–Baxter quantum equation" [5]

$$\mathscr{R}^{12}\mathscr{R}^{13}\mathscr{R}^{23} = \mathscr{R}^{23}\mathscr{R}^{13}\mathscr{R}^{12},$$

where by $\mathscr{R}^{12}$ (for example) we mean $FE^{-1} \otimes EF^{-1} \otimes 1$.

Returning to consider U, we see that deformations obtained by "twisting" in this way have a nice feature: If I is some two-sided Hopf ideal in U, then the two-sided ideal $I[[h]] = \{\sum_{i=0}^{\infty} \xi_i h^i \mid \xi_i \in I\ \forall i\}$ is a quasi-Hopf ideal in $U^{E,F}$. By this we mean that

(a) $\Delta_{E,F}(I[[h]]) \subseteq I[[h]] \mathbin{\hat{\otimes}_{\mathbb{Z}[[h]]}} U^{E,F} + U^{E,F} \mathbin{\hat{\otimes}_{\mathbb{Z}[[h]]}} I[[h]]$,

and

(b) $S(I[[h]]) \subseteq I[[h]]$.

This should be clear by inspecting the formulas in the above proof. Thus we have that $U^{E,F}/I[[h]]$ is a quasi-Hopf algebra—whose Drinfel'd associator, unit constraints, and antipode constraints are those obtained from $U^{E,F}$. The "flatness" property of deformations allows one to conclude that

THEOREM 3.0.25. *$U^{E,F}/I[[h]]$ is a quasi-Hopf algebra deformation of U/I.*

Using this theorem, and making particular choices of the curves E and F and the ideal I, we get some rather interesting non-coassociative bialgabras, the study of which we defer to [6].

In conclusion, we make some remarks about the graded cases. Here, we add the requirement that E and F satisfy the condition degree$(E_i) =$ degree$(F_i) =$ degree$(X_i) = Ki$ for every i. Then one can calculate that the coefficients of h^n in $\Delta_{E,F}(X_m)$ have degree $Km + Kn$; thus it is natural to assign the parameter h the degree $-K$. We would like to think of $U[[h]]$ as the completion of the polynomial ring $U[h]$ with respect to the degree (in h) filtration. The ideal I of U above is replaced by a graded ideal I_0 in U. In $U^{E,F}/I_0[[h]]$, if ξ is a homogeneous element of U/I_0 of degree r, the coefficient of h^n in $\Delta_{E,F}(\xi)$ has degree $Kn + r$.

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